

Russell Corbett

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Experience

Nasdaq Verafin

May 2024 – Aug 2024, Jan 2025 – April 2025

Software Developer Intern

- Debugged and troubleshooted software issues directly with users, improving customer experience.
- Modified database import parameters to optimize data transmission into the Verafin application.

Angler Solutions

Sep 2023 – Dec 2023

Co-op Engineering Student

- Configured network systems for ten partner companies, including securing video transmission, peer-to-peer device communications, and deploying live streams to an online audience.
- Developed and led a team of three following agile practices to create a wind farm web application to analyze sensor readings to determine optimal turbine location using Python.

Bell Canada Enterprises

Jan 2023 – Apr 2023

System Analyst

- Developed a data analysis automation system using Python to reduce manual work by 99.7%.
- Implemented new automation creation processes for future co-op students to follow, unifying design and format.

Avalon Controls

May 2022 – Aug 2022

Engineering Co-op Intern

- Architected and implemented a new communications and networking system, increasing the speed 100x.
- Developed a deployment plan for future system expansion, ensuring scalability and maintainability.

Eastern Edge Robotics

Oct 2021 – Present

Chief Operating Officer and Former Software Lead (Extra Curricular)

- Architected and developed a competitive underwater ROV as part of a robotics design team.
- Developed a custom control system for the ROV, including a GUI for diagnostics and improved control for the pilot.
- Integrated brushless motor thrusters, servos, lighting rigs, network communications, and onboard tooling with ROV control systems using Python and ROS2.
- Placed 1st in Engineering Presentation at the 2022 and 2023 MATE ROV World Championship.

Paradigm Engineering

October 2022 – April 2024

Path Planning Lead (Extra Curricular)

- Developed a path-planning algorithm for a high-speed vehicle to navigate a dynamic environment.
- Designed a vehicle and pit crew communication system to ensure real-time updates using Python and UDP.
- Tested the algorithm and system in a ROS2 simulation environment, ensuring robustness and reliability.

Education

Memorial University of Newfoundland

St. John's, NL, Canada

Bachelor of Engineering - Computer Engineering

Class of 2026

Engineering Society Director of Communications (2023-2025), Relay for Life Tech Committee Lead

Skills

Languages: JavaScript, TypeScript, C/C++, Python, C#, HTML, CSS

Frontend: ReactJS, Astro, NextJS, RemixJS, TailwindCSS

Backend: NodeJS, ExpressJS

Embedded: Arduino, Raspberry Pi, ESP32, ROS2

Tools: Git, GitHub Actions, Docker, Grafana, AWS, Digital Ocean, Proxmox VE

Projects

Corbett Homelab

- Architected and implemented a server environment using Proxmox VE and Docker to host various services and apps.
- Deployed control and communication systems using MQTT, Zigbee, Z-Wave, and Bluetooth protocols.
- Configured and maintained over 2200 smart home entities, including lights, sensors, and cameras.
- Reduced home electricity consumption by 40% by automating lighting and heating systems.